

10-Mark Questions & Answers

Q1. Explain Computer Architecture vs Computer Organization in detail with a suitable diagram and real-world example (e.g., same architecture, different organization).

1. Fundamental Distinction

* **Computer Architecture:** Refers to those attributes of a system visible to a programmer, which directly impact the logical execution of a program. It defines the abstract software-hardware interface (ISA - Instruction Set Architecture).

* **Key**
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2. Conceptual Diagram

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3. Comprehensive Comparison Table

| Feature | Computer Architecture | Computer Organization |

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4. Real-World Example: Same Architecture, Different Organization

The x86 Architecture (originally introduced with Intel 8086) has remained architecturally compatible for decades:

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Q2. Describe the Von Neumann model in detail - architecture diagram, components, key features, instruction cycle (Fetch-Decode-Execute-Store), advantages and disadvantages.

1. Architecture Diagram

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2. Major Components

1. Control Unit (CU): The central manager. Fetches instructions, decodes opcodes, and issues timing/control signals to all internal components.

2. Arith

3. Key Features

* **Stored Program Concept:** Code and data share a single memory space and bus.

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Program

4. Instruction Cycle Flow

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Fetch In

5. Advantages and Disadvantages

* **Advantages:**

* Hard

Q3. Compare Von Neumann and Harvard architecture in detail with diagrams and a comparison table, and discuss which is better for modern processors and why.

1. Architectural Diagrams

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HARVARD ARCHITECTURE:

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2. Detailed Comparison Table

| Feature | Von Neumann Architecture | Harvard Architecture |

| :-:- | :-:-

3. Which is Better for Modern Processors and Why?

Modern high-performance processors (like Intel Core, AMD Ryzen, ARM Cortex-A) use a Modified Harvard Architecture internally, combined with a Von Neumann Architecture externally:

* **Internal Level (Harvard Concept for Speed):**

* Proc

Conclusion: The Modified Harvard Architecture is superior for modern high-speed processors because it delivers the ultra-fast execution of dual-bus Harvard caching inside the CPU while preserving the cheap, flexible unified RAM of Von Neumann on the motherboard.

Q4. Describe the architecture of the 8086 microprocessor in detail with a block diagram - explain BIU, EU, registers, operating modes, and memory organization.

1. Internal Block Diagram

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2. Functional Units: BIU vs EU

* **Bus Interface Unit (BIU):**

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3. Register Organization

1. General-Purpose Registers (16-bit / 8-bit pairs):

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4. Memory Organization (Segmentation)

* 20-bit address bus gives **1 MB (2^{20} bytes)** total memory space.

* Mem

5. Operating Modes

* **Minimum Mode (MN/MX' = 5V):** Single-processor mode; 8086 generates all control signals directly.

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Control

Q5. Explain all the performance measures of computer architecture with formulas, and solve a numerical problem using them.

1. Performance Measures & Formulas

1. CPU Execution Time (T_{cpu}): Total time taken to execute a program.

Equation:

T_{cpu}

Equation:

T_{cpu}

2. Performance (P): Inverse of execution time.

Equation:

$P = \frac{1}{T_{\text{cpu}}}$

3. Clock Rate (f): Clock frequency of the processor oscillator.

Equation:

$$f = \frac{1}{T}$$

4. CPI (Cycles Per Instruction): Average clock cycles required per instruction.

Equation:

$$CPI = \frac{\text{Total Cycles}}{\text{Total Instructions}}$$

5. MIPS (Millions of Instructions Per Second):

Equation:

$$MIPS = \frac{\text{Instructions}}{\text{Seconds} \times 10^6}$$

6. MFLOPS (Millions of Floating-Point Operations Per Second):

Equation:

$$MFLOPS = \frac{\text{Floating-Point Operations}}{\text{Seconds} \times 10^6}$$

7. Throughput: Number of complete execution tasks finished per unit time.

8. Latency

2. Numerical Problem & Solution

Problem: A 3.0 GHz processor executes a benchmark program with 20 million instructions. The instruction mix and CPI breakdown are as follows:

* Arith

Calculate:

1. Aver

Solution:

* **Step 1: Calculate Average CPI**

Equation:

$$\text{Average CPI} = \frac{\sum (\text{CPI}_i \times \text{Instructions}_i)}{\text{Total Instructions}}$$

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Equation:

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* **Step 2: Calculate CPU Execution Time**

* **Ins

Equation:

Equation:

* **Step 3: Calculate MIPS Rating**

Equation:

Q6. Explain the complete instruction set of 8086 - arithmetic, data transfer, control transfer, logical, and shift/rotate instructions - with syntax and examples for each category.

1. Data Transfer Instructions

Used to transfer binary data between registers, memory, and I/O ports.

2. Arithmetic Instructions

Perform binary addition, subtraction, multiplication, and division.

3. Logical Instructions

Perform bitwise logical operations and comparisons.

4. Shift and Rotate Instructions

Bitwise shifting and rotation of operand contents.

5. Control Transfer Instructions

Alter instruction execution flow.

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Q7. Explain all addressing modes of the 8086 microprocessor with syntax, effective address formula, and a worked numerical example for each mode.

Given Register & Memory State for Numerical Examples

Let: `DS = 2000H`, `SS = 3000H`, `BX = 1000H`, `BP = 2000H`, `SI = 0100H`, `DI = 0200H`, `Displacement = 0050H`.

1. Immediate Addressing Mode

* **Concept:** Operand is constant data given inside the instruction.

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2. Register Addressing Mode

* **Concept:** Operand is inside a register.

* **Syn

3. Direct Addressing Mode

* **Concept:** 16-bit offset is explicitly given in the instruction.

* **Syn

4. Register Indirect Addressing Mode

* **Concept:** Offset is held in a base/index register (`BX`, `BP`, `SI`, `DI`).

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5. Based Addressing Mode

* **Concept:** EA is base register (`BX` or `BP`) plus displacement.

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6. Indexed Addressing Mode

* **Concept:** EA is index register (`SI` or `DI`) plus displacement.

* **Syn

7. Based-Indexed Addressing Mode

* **Concept:** EA is the sum of a base register, index register, and optional displacement.

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Q8. Draw and explain the block diagram of the 8086 microprocessor, labeling all registers, ALU, flag register, instruction queue, and explain how pipelining (BIU + EU) improves performance.

1. Detailed Block Diagram

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2. Components Explanation

* **BIU:** Prefetches instructions, calculates 20-bit physical address, interfaces with memory bus.

* **EU**

3. Pipelining in 8086 (BIU + EU Parallelism)

* **Non-Pipelined System:** CPU fetches instruction 1 $\rightarrow$ executes instruction 1 $\rightarrow$ fetches instruction 2 $\rightarrow$ executes instruction 2. Memory bus is idle during execution, CPU is idle during fetch.

* **8086**

Q9. Numerical: Given a program's instruction mix with respective clock cycles, calculate total CPU time, CPI, and MIPS.

Problem Statement

A 2.5 GHz (2.5×10^9 Hz) processor executes a program with a total of 40 million instructions (40×10^6). The program consists of the following instruction mix:

1. Arithmetic & Logical Instructions: 45% of total instructions, requiring 2 clock cycles per instruction.

2. Data

Calculate:

1. Total

Solution

Step 1: Calculate Instruction Counts per Category

* Total

Step 2: Calculate Total Clock Cycles

Equation:

Total C

Equation:

Total C

Equation:

Total C

Step 3: Calculate Average CPI

Equation:

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Equation:

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Step 4: Calculate CPU Execution Time

Equation:

CPU Ti

Equation:

T_{cpu}

Step 5: Calculate MIPS Rating

Equation:

MIPS =

Equation:

MIPS =

Alternatively:

Equation:

MIPS =
